Plasmon-Pole Models: Asymptotic Exactness in the Long-Wavelength Limit and its Breakdown in a Finite Plane-Wave Basis

1 Setup: $\mathbf{G_0W_0}$ self-energy

We work with a uniform electron gas (UEG) at the Gamma point with N_e electrons ($N_{\rm occ} = N_e/2$ occupied orbitals) in a plane-wave basis of size M, due to truncation at energy $E_{\rm cut}$. The correlation part of the $\mathrm{G_0W_0}$ self-energy evaluated on the real frequency axis is

$$\Sigma^c(\mathbf{k},\omega) = \frac{\mathrm{i}}{2\pi}\sum_{\mathbf{q}}\int_{-\infty}^{\infty} d\omega'\, G_0(\mathbf{k}-\mathbf{q},\,\omega-\omega')\, W_c(\mathbf{q},\omega'), \tag{1}$$

where $G_0(\mathbf{k},\omega) = f_{\mathbf{k}}/(\omega-\varepsilon_{\mathbf{k}}-\mathrm{i}\eta) + (1-f_{\mathbf{k}})/(\omega-\varepsilon_{\mathbf{k}}+\mathrm{i}\eta)$ and $W_c = W - v$ is the correlation part of the screened Coulomb interaction.

Closing the ω'-contour and using the Lehmann representation of W_c gives

$$\Sigma^c(\mathbf{k},\omega) = \sum_{\mathbf{q}}\sum_{\mu}\left[\frac{(1-f_{\mathbf{k}-\mathbf{q}})\,|\mathcal{Z}^{\mathbf{q}}_\mu|^2}{\omega-\varepsilon_{\mathbf{k}-\mathbf{q}}-\Omega^{\mathbf{q}}_\mu+\mathrm{i}\eta} + \frac{f_{\mathbf{k}-\mathbf{q}}\,|\mathcal{Z}^{\mathbf{q}}_\mu|^2}{\omega-\varepsilon_{\mathbf{k}-\mathbf{q}}+\Omega^{\mathbf{q}}_\mu-\mathrm{i}\eta}\right], \tag{2}$$

where $\Omega^{\mathbf{q}}_\mu > 0$ are the RPA charged excitation energies for momentum transfer $\mathbf{q}$ and $|\mathcal{Z}^{\mathbf{q}}_\mu|^2$ are the corresponding transition densities. In a finite basis with $N_{\rm occ}$ occupied and $N_{\rm vir}$ virtual orbitals, the index μ runs over at most $N_{\rm occ}\times N_{\rm vir}$ poles per $\mathbf{q}$.

2 Spectral density and its moments

Using the relation between the transition densities and the bare oscillator strengths as $\left|\mathcal{Z}^{\mathbf{q}}_\mu\right|^2 = v(\mathbf{q})\left|B^{\mathbf{q}}_\mu\right|^2$, define the *spectral density of the inverse dielectric function*

$$B(\mathbf{q},\nu) \equiv -\frac{1}{\pi}\,\mathrm{Im}\big[\varepsilon^{-1}(\mathbf{q},\nu)-1\big] = \sum_{\mu}|\mathcal{B}^{\mathbf{q}}_\mu|^2\,\delta(\nu-\Omega^{\mathbf{q}}_\mu), \qquad \nu>0, \tag{3}$$

implying the Kramers–Kronig form

$$\varepsilon^{-1}(\mathbf{q},\omega) - 1 = \int_0^\infty d\nu\,\frac{2\nu\,B(\mathbf{q},\nu)}{\omega^2-\nu^2+\mathrm{i}\eta}. \tag{4}$$

Define the *spectral moments*

$$M_n(\mathbf{q}) \equiv \int_0^\infty d\nu\,\nu^n\,B(\mathbf{q},\nu) = \sum_\mu |\mathcal{B}^{\mathbf{q}}_\mu|^2\,(\Omega^{\mathbf{q}}_\mu)^n. \tag{5}$$

The two moments that are cheaply accessible without diagonalising the full RPA matrix are the static limit M_{-1}

$$-2M_{-1}(\mathbf{q}) = -2\int_0^\infty d\nu\, \nu^{-1} B(\mathbf{q},\nu) \tag{6}$$

$$= \int_0^\infty d\nu\, \frac{2\nu\, B(\mathbf{q},\nu)}{-\nu^2 + \mathrm{i}\eta} \tag{7}$$

$$= \varepsilon^{-1}(\mathbf{q},0) - 1, \tag{8}$$

and the f-sum rule M_1 (which is a statement that the first spectral moment is related to the finite-basis plasmon frequency)

$$M_1(\mathbf{q}) = \int_0^\infty d\nu\, \nu\, B(\mathbf{q},\nu) = \sum_\mu |\mathcal{B}_\mu^{\mathbf{q}}|^2 \Omega_\mu^{\mathbf{q}} = \frac{1}{2}\omega_p^2(\mathbf{q}). \tag{9}$$

2.1 Finite-basis plasmon frequency in a finite basis vs. the complete-basis limit

The noninteracting density response at momentum transfer $\mathbf{q}$ is

$$\chi_0(\mathbf{q},\omega) = \frac{1}{V}\sum_{\substack{i\in\mathrm{occ}\\ a\in\mathrm{vir}}} \left|\langle a|e^{\mathrm{i}\mathbf{q}\cdot\hat{r}}|i\rangle\right|^2 \left[\frac{1}{\omega - (\varepsilon_a - \varepsilon_i) + \mathrm{i}\eta} - \frac{1}{\omega + (\varepsilon_a - \varepsilon_i) - \mathrm{i}\eta}\right] \tag{10}$$

$$= \frac{1}{V}\sum_{\substack{i\in\mathrm{occ}\\ a\in\mathrm{vir}}} \left|\langle a|e^{\mathrm{i}\mathbf{q}\cdot\hat{r}}|i\rangle\right|^2 \frac{2(\varepsilon_a - \varepsilon_i)}{\omega^2 - (\varepsilon_a - \varepsilon_i)^2 + \mathrm{i}\eta} \tag{11}$$

$$\approx \frac{2}{V\omega^2}\sum_{\substack{i\in\mathrm{occ}\\ a\in\mathrm{vir}}} \left|\langle a|e^{\mathrm{i}\mathbf{q}\cdot\hat{r}}|i\rangle\right|^2 (\varepsilon_a - \varepsilon_i) \qquad |\omega| \gg \varepsilon_a - \varepsilon_i. \tag{12}$$

Using

$$\varepsilon^{-1}(\mathbf{q},\omega) - 1 \equiv \frac{v(\mathbf{q})\chi_0(\mathbf{q},\omega)}{1 - v(\mathbf{q})\chi_0(\mathbf{q},\omega)} = v(\mathbf{q})\chi_0(\mathbf{q},\omega) + \mathcal{O}\big((v\chi_0)^2\big) \tag{13}$$

$$\approx v(\mathbf{q})\chi_0(\mathbf{q},\omega) \qquad |\omega| \to \infty \tag{14}$$

$$= \frac{2v(\mathbf{q})}{V\omega^2}\sum_{\substack{i\in\mathrm{occ}\\ a\in\mathrm{vir}}} \left|\langle a|e^{\mathrm{i}\mathbf{q}\cdot\hat{r}}|i\rangle\right|^2 (\varepsilon_a - \varepsilon_i). \tag{15}$$

On the other hand, the spectral representation gives

$$\varepsilon^{-1}(\mathbf{q},\omega) - 1 = \int_0^\infty d\nu\, \frac{2\nu B(\mathbf{q},\nu)}{\omega^2 - \nu^2 + \mathrm{i}\eta} \tag{16}$$

$$\approx \frac{2}{\omega^2}\int_0^\infty d\nu\, \nu B(\mathbf{q},\nu) \qquad |\omega| \to \infty \tag{17}$$

$$= \frac{2M_1(\mathbf{q})}{\omega^2}. \tag{18}$$

Equating the coefficients of $1/\omega^2$ in Eqs. (15) and (18) gives

$$M_1(\mathbf{q}) = \frac{v(\mathbf{q})}{V} \sum_{\substack{i\in\text{occ}\\ a\in\text{vir}}} \left|\langle a|e^{\mathrm{i}\mathbf{q}\cdot\hat{r}}|i\rangle\right|^2 (\varepsilon_a - \varepsilon_i) = \frac{1}{2}\omega_p^2(\mathbf{q}), \tag{19}$$

Now we will specify to the uniform electron gas. In the finite basis, we have

$$\omega_p^2(\mathbf{q}) = \frac{2v(\mathbf{q})}{V} \sum_{\substack{i\in\text{occ}\\ a\in\text{vir}}} \left|\delta_{\mathbf{G}_a,\mathbf{G}_i+\mathbf{q}}\right|^2 (\varepsilon_a - \varepsilon_i), \tag{20}$$

$$= \frac{2v(\mathbf{q})}{V} \sum_{\substack{i\in\text{occ}\\ \mathbf{G}_i+\mathbf{q}\in\text{vir}}} (\varepsilon_{\mathbf{G}_i+\mathbf{q}} - \varepsilon_{\mathbf{G}_i}). \tag{21}$$

whereas in the CBS limit, the sum over virtual states becomes complete and we instead get

$$\sum_{a\in\text{vir}} \left|\langle a|e^{\mathrm{i}\mathbf{q}\cdot\hat{r}}|i\rangle\right|^2 (\varepsilon_a - \varepsilon_i) = \frac{q^2}{2} \tag{22}$$

$$\implies \sum_{i\in\text{occ}} \frac{q^2}{2} = \frac{N_e q^2}{2}. \tag{23}$$

Substituting this CBS value into Eq. (19) gives the Sommerfeld bulk plasma frequency,

$$\omega_p^2 = \frac{2}{V}\frac{4\pi}{q^2}\frac{N_e q^2}{2} \tag{24}$$

$$= 4\pi\frac{N_e}{V} \tag{25}$$

$$= 4\pi n. \tag{26}$$

So ω_p takes on a different value in a finite basis versus the CBS.

3 The Hybertsen–Louie plasmon-pole model

The HL PPM replaces the full spectral density of eq. (3) with a single delta peak,

$$B^{\text{PPM}}(\mathbf{q},\nu) = \tilde{A}(\mathbf{q})\,\delta(\nu - \tilde{\omega}(\mathbf{q})), \tag{27}$$

parametrised by two unknowns $\{\tilde{A},\tilde{\omega}\}$ per $\mathbf{q}$, fixed by matching the two accessible moments (8)–(9):

$$M_{-1}^{\text{PPM}} = \frac{\tilde{A}}{\tilde{\omega}} = M_{-1} \quad\Longrightarrow\quad \tilde{A} = M_{-1}\,\tilde{\omega}, \tag{28}$$

$$M_1^{\text{PPM}} = \tilde{A}\,\tilde{\omega} = M_1 \quad\Longrightarrow\quad \tilde{A}\,\tilde{\omega} = \tfrac{1}{2}\omega_p^2. \tag{29}$$

Dividing (29) by (28) gives the *HL pole frequency*

$$\boxed{\tilde{\omega}^2(\mathbf{q}) = \frac{M_1(\mathbf{q})}{M_{-1}(\mathbf{q})} = \frac{\omega_p^2(\mathbf{q})/2}{-\big(\varepsilon^{-1}(\mathbf{q},0)-1\big)/2}.} \tag{30}$$

Substituting back into Eq. (2), the PPM self-energy becomes

$$\Sigma_c^{\text{PPM}}(\mathbf{k},\omega) = \sum_{\mathbf{q}} \left[\frac{(1-f_{\mathbf{k}-\mathbf{q}})\,\tilde{A}(\mathbf{q})}{\omega - \varepsilon_{\mathbf{k}-\mathbf{q}} - \tilde{\omega}(\mathbf{q}) + \mathrm{i}\eta} + \frac{f_{\mathbf{k}-\mathbf{q}}\,\tilde{A}(\mathbf{q})}{\omega - \varepsilon_{\mathbf{k}-\mathbf{q}} + \tilde{\omega}(\mathbf{q}) - \mathrm{i}\eta}\right]. \tag{31}$$

4 Exactness condition

Comparing Eq. (2) with Eq. (31) for the 2h1p sector, we see that the PPM is exact if and only if, for every $\mathbf{q}$ and every ω,

$$\sum_{\mu} \frac{v(\mathbf{q})|\mathcal{B}_{\mu}^{\mathbf{q}}|^2}{\omega - \varepsilon_{\mathbf{k}-\mathbf{q}} - \Omega_{\mu}^{\mathbf{q}}} = \frac{M_{-1}(\mathbf{q})\,\tilde{\omega}(\mathbf{q})}{\omega - \varepsilon_{\mathbf{k}-\mathbf{q}} - \tilde{\omega}(\mathbf{q})}. \tag{32}$$

and we know

$$W \equiv M_0(\mathbf{q}) = \sum_{\mu} |\mathcal{B}_{\mu}^{\mathbf{q}}|^2 \tag{33}$$

Now, defining $x \equiv \omega - \varepsilon_{\mathbf{k}-\mathbf{q}}$, $w_\mu \equiv |\mathcal{B}_{\mu}^{\mathbf{q}}|^2$ and $\nu_\mu \equiv \Omega_{\mu}^{\mathbf{q}}$, we can rewrite the left side by splitting each resolvent:

$$\sum_{\mu} \frac{w_\mu}{x - \nu_\mu} = \frac{W}{x - \tilde{\omega}} + \sum_{\mu} w_\mu \frac{\tilde{\omega} - \nu_\mu}{(x - \nu_\mu)(x - \tilde{\omega})}. \tag{34}$$

Expanding $(x - \nu_\mu)^{-1}$ about $(x - \tilde{\omega})^{-1}$ in powers of $\delta\nu_\mu \equiv \nu_\mu - \tilde{\omega}$ yields

$$\sum_{\mu} \frac{w_\mu}{x - \nu_\mu} = \frac{W}{x - \tilde{\omega}} \left[1 + \frac{\langle (\delta\nu)^2 \rangle_w}{(x - \tilde{\omega})^2} + \frac{\langle (\delta\nu)^3 \rangle_w}{(x - \tilde{\omega})^3} + \cdots \right], \tag{35}$$

where $\langle f \rangle_w \equiv \sum_\mu w_\mu f(\nu_\mu)/W$ and we used the fact that $\langle \delta\nu \rangle_w = 0$ by construction of $\tilde{\omega}$ as the w-weighted mean.

For Eq. (32) to hold, every correction in Eq. (35) must vanish. Provided $|x - \tilde{\omega}|$ is not too small, the dominant correction involves the *spectral variance* $\mathrm{Var}_w[\Omega^{\mathbf{q}}] \equiv \langle (\delta\nu)^2 \rangle_w$, giving the leading PPM error

$$\Sigma_c(\mathbf{k}, \omega) - \Sigma_c^{\mathrm{PPM}}(\mathbf{k}, \omega) \;\approx\; \sum_{\mathbf{q}} (1 - f_{\mathbf{k}-\mathbf{q}}) \frac{W(\mathbf{q})\ \mathrm{Var}_w[\Omega^{\mathbf{q}}]}{(\omega - \varepsilon_{\mathbf{k}-\mathbf{q}} - \tilde{\omega}(\mathbf{q}))^3} + \cdots \tag{36}$$

Necessary and sufficient condition for PPM exactness. *The HL PPM reproduces the exact GW self-energy if and only if*

$$\mathrm{Var}_w[\Omega^{\mathbf{q}}] = 0 \quad \text{for every } \mathbf{q}, \tag{37}$$

i.e. the RPA spectral function $B(\mathbf{q}, \cdot)$ is a single delta peak concentrated at one excitation energy.

5 Why the PPM is exact in the thermodynamic limit

In the TDL ($N_e \to \infty$, $V \to \infty$, $n = N_e/V$ fixed) the RPA spectral function of the UEG separates into *one* sharp collective (plasmon) mode and an incoherent particle-hole (p-h) continuum:

$$B^{\mathrm{TDL}}(\mathbf{q}, \nu) = Z_p(\mathbf{q})\, \delta\big(\nu - \omega_p(\mathbf{q})\big) + B_{\mathrm{ph}}(\mathbf{q}, \nu)\, \Theta\big(\nu - \nu_{\mathrm{ph}}^{\min}(\mathbf{q})\big), \tag{38}$$

where $\omega_p(\mathbf{q}) = \sqrt{4\pi n q^2/m}$ is the long-wavelength plasmon and $\nu_{\mathrm{ph}}^{\min}(\mathbf{q}) = q^2/2 + qk_F$ is the lower edge of the p-h continuum (k_F the Fermi momentum).

For $q \ll q_c$ (the Landau cutoff), the plasmon lies *below* the continuum and exhausts nearly all the f-sum weight:

$$M_1^{\mathrm{TDL}}(\mathbf{q}) = Z_p\,\omega_p + \int_{\nu_{\mathrm{ph}}^{\min}}^{\infty} d\nu\,\nu\,B_{\mathrm{ph}}(\mathbf{q},\nu) \;\xrightarrow{q\to 0}\; Z_p\,\omega_p. \tag{39}$$

As the p-h continuum weight $\int B_{\mathrm{ph}} \to 0$, the spectral density converges to a single delta function, so

$$\mathrm{Var}_w^{\mathrm{TDL}}[\Omega^{\mathbf{q}}] \;\propto\; \frac{\int B_{\mathrm{ph}}\,(\nu-\omega_p)^2\,d\nu}{\int B_{\mathrm{ph}}\,d\nu + Z_p} \;\to\; 0 \quad \text{as } q\to 0,\; N_k\to\infty. \tag{40}$$

The correction (36) therefore vanishes and the PPM becomes exact in the TDL for small-q momentum transfers.

6 Finite-basis breakdown

On a finite k-point mesh the RPA spectrum at each $\mathbf{q}$ is discrete with at most $N_{\mathrm{occ}} \times N_{\mathrm{vir}}$ poles $\{\Omega_\mu^{\mathbf{q}}\}$. Since:

$$\mathrm{Var}_w[\Omega^{\mathbf{q}}] = \frac{M_0 M_2 - M_1^2}{M_0^2} \;\geq\; 0, \tag{41}$$

with equality only when all poles are identical. By Jensen's inequality for the convex function $\phi(x) = x^2$:

$$\mathrm{Var}_w[\Omega^{\mathbf{q}}] = 0 \iff \Omega_\mu^{\mathbf{q}} = \text{const for all } \mu. \tag{42}$$

In a finite PW basis this is never satisfied: the particle-hole transition energies $\varepsilon_a - \varepsilon_i$ span a range of order the bandwidth, so different pairs (i,a) at momentum transfer $\mathbf{q}$ produce inequivalent poles, giving $\mathrm{Var}_w > 0$.

A second, independent source of error is *basis incompleteness*. The finite-basis f-sum moment (**??**) omits transitions to states above E_{cut}:

$$M_1^{\mathrm{finite}}(\mathbf{q}) = \sum_{\substack{i\in\mathrm{occ}\\ a\in\mathrm{vir}}} |\langle a|e^{\mathrm{i}\mathbf{q}\cdot\hat{r}}|i\rangle|^2(\varepsilon_a-\varepsilon_i) \;<\; M_1^{\mathrm{TDL}}(\mathbf{q}), \tag{43}$$

which shifts $\tilde{\omega}(\mathbf{q})$ downward (Eq. (30)) and introduces a systematic bias in the pole frequency, independent of the multi-pole variance effect.

Combining both contributions, the total PPM error is

$$\Sigma_c - \Sigma_c^{\mathrm{PPM}} \;=\; \underbrace{\sum_{\mathbf{q}} \frac{(1-f_{p-q})\,W(\mathbf{q})\;\mathrm{Var}_w[\Omega^{\mathbf{q}}]}{(\omega-\varepsilon_{p-q}-\tilde{\omega})^3}}_{\text{multi-pole residual}} \;+\; \underbrace{\mathcal{O}\!\left(\frac{M_1^{\mathrm{TDL}}-M_1^{\mathrm{finite}}}{(\omega-\varepsilon_{p-q})^2}\right)}_{\text{basis-incompleteness shift}} + \cdots \tag{44}$$

Both terms vanish as $N_k, E_{\mathrm{cut}} \to \infty$, but are strictly nonzero for any finite discretisation, consistent with the $\sim 10^{-7}$ eV discrepancies observed numerically.